

Real Estate Voice Agent

End-to-end n8n backend for a Retell phone-call voice agent — technical & process documentation

AI Automation & n8n Workflow Developer

n8n

Retell

Google Sheets

Gmail

This report documents the n8n backend that powers a Retell phone voice agent for a real estate office (built for the fictional Crestline Realty Group as a portfolio example). Callers can search available listings, list a property they want to sell or rent, book a viewing, ask about a specific property, or escalate to a human — entirely by voice — while n8n validates every request, keeps a Google Sheets record, and keeps the office notified by email.

1. Executive Summary

A real estate office fields constant phone traffic: buyers asking what's available, owners wanting to list a property, people booking viewings, and callers who just want a human. This backend gives a Retell voice agent five callable tools that handle exactly these requests, each backed by validation logic so the agent searches real inventory, never double-books a viewing slot, and always routes sensitive conversations (financing, negotiation) to a person.

Metric	Detail
Trigger	5 webhooks called directly by Retell during a live phone call
Voice platform	Retell — the caller talks to an AI agent, which calls these tools mid-conversation
Data store	Google Sheets — Listings, Viewings and Escalations tabs
Office notifications	Gmail — sent on every new listing submission, booked viewing and escalation
New listings	Saved as pending_review — never published automatically, staff approve first
Viewing conflicts	A property/date/time slot already booked is rejected before it reaches the sheet

2. Business Value

- Round-the-clock lead capture — buyers get instant answers on availability and pricing any time they call, not just office hours.
- New inventory funnel — owners and landlords can list a property over the phone; it's queued for staff approval instead of requiring a form or an in-person visit.
- No double-booked viewings — every visit request is checked against existing bookings for that exact property, date and time before it's confirmed.
- Nothing sensitive handled by AI — financing questions, negotiation, or anything outside the agent's scope is escalated straight to a human with full context.
- Full visibility for the office — every listing, viewing and escalation is logged to Sheets and emailed immediately, with a direct call link on urgent escalations.

3. Architecture Overview

Retell's voice agent is configured with five tools, each pointing at its own n8n webhook. During a call, the agent decides which tool to call based on what the caller is asking for.

Tool / Webhook	Purpose
check_property_availability	Searches Listings by location, type, bedrooms, budget and sale/rent, returning up to 5 matches
list_property	Saves a caller's property as pending_review and emails the office for approval
book_property_visit	Checks the requested slot is free, logs the viewing, emails the assigned agent
get_property_details	Looks up one listing by property ID or address and reads back price, size, bedrooms and description
request_human_callback	Logs an escalation (financing, negotiation, anything out of scope) and alerts the office

4. Tool — check_property_availability

1 Capture the search criteria

Location, property type, bedroom count, maximum budget and listing type (buy/rent) are pulled from the caller's request.

2 Read the inventory

All rows are pulled from the Listings sheet so they can be filtered against the caller's criteria.

3 Filter intelligently

Listings already sold, rented, or still pending review are excluded automatically. Location matching is partial (so “downtown” matches “downtown Austin”), bedroom count is a minimum rather than an exact match, and “buy/purchase” and “lease/renting” are normalised to sale/rent so casual phrasing still matches correctly.

4 Reply in natural language

Up to 5 matching properties are summarised into a sentence Retell can speak directly to the caller (property ID, bedrooms, type, location and price), or a friendly “nothing found, want to adjust budget or location?” prompt when there are no matches.

5. Tool — list_property

1 Capture the submission

Owner name, phone number, property type, location, size, price, bedrooms, description and listing type (sale/rent) are collected from the caller.

2 Generate a reference & save

A unique property ID is generated and the listing is appended to the Listings sheet with status pending_review — it is never published or made searchable automatically.

3 Notify the office

A formatted email goes to the office with every submitted detail (owner, contact, type, location, size, price, bedrooms and description) so staff can review and approve it.

4 Confirm back to the caller

The caller is told their property was submitted for review under their reference number and that the team will list it shortly.

6. Tool — book_property_visit

- 1 Capture the request**
Customer name, phone number, property ID, and the requested date and time are collected.
- 2 Check the slot**
Existing viewings for that exact property are pulled, dates and times are normalised into a consistent format, and the requested slot is checked against confirmed (non-cancelled) bookings for a conflict.
- 3 Available — book it**
The viewing is appended to the Viewings sheet, the assigned agent is emailed the customer's details and the requested time, and the caller receives a confirmation naming the property, date and time.
- 4 Unavailable — offer an alternative**
If the slot is already taken, no booking is made; the caller is told that time is unavailable and asked if another time would work.

7. Tool — get_property_details

- 1 Capture the lookup**
The caller's property ID (preferred) or a location/address is captured from the request.
- 2 Search the listings**
The Listings sheet is read and matched first by exact property ID, falling back to a partial location match if no ID was given.
- 3 Speak back the details**
When a match is found, the agent reads back bedrooms, property type, location, size, price, listing type and the full description in one natural sentence. When nothing matches, the caller is asked to confirm the property ID or address instead of getting a dead end.

8. Tool — request_human_callback

1

Capture the escalation

Customer name, phone number and the reason they need a human (financing questions, negotiation, or anything the agent shouldn't handle) are recorded.

2

Log & alert the office

The request is appended to a dedicated Escalations tab, and the office receives an urgent-styled email with the customer's details, the reason, and a one-tap call link.

3

Confirm back to the caller

The caller is told the request has been flagged for the team and that someone will call them back shortly — this tool never transfers the live call, it only queues a follow-up.

9. Node-by-Node Map

Tool	Key nodes
check_property_availability	Check Availability Webhook → Normalize Search Criteria → Read Listings (Search) → Filter Matching Properties → Respond Availability Search
list_property	List Property Webhook → Normalize Listing Data → Append Listing to Sheet → Build Listing Confirmation / Send Team New Listing Email → Respond List Property
book_property_visit	Book Visit Webhook → Normalize Visit Request → Read Viewings (Slot Check) → Check Visit Slot → Is Visit Slot Available → Append Visit to Sheet / Build Visit Unavailable Message → Build Visit Confirmation / Send Agent New Visit Email → Respond Book Visit
get_property_details	Property Details Webhook → Normalize Details Request → Read Listings (Details) → Find Property Details → Respond Property Details
request_human_callback	Escalation Webhook → Normalize Escalation Data → Append Escalation to Sheet → Send Office Escalation Email / Build Escalation Response → Respond Escalation

10. Reliability & Security Design

- **No auto-published listings** — every submitted property lands as pending_review; nothing goes live without staff approving it.
- **Server-side conflict checking** — viewing availability is verified against the sheet at booking time, per property, date and time, so double-bookings can't slip through.
- **Tolerant matching, safe defaults** — location, bedroom and listing-type matching handles natural phrasing (partial locations, “buy” vs “purchase”) without over-matching sold or pending listings.
- **Clear escalation path** — financing, negotiation and anything outside the agent's scope has a dedicated tool that guarantees a human follow-up rather than an improvised AI answer.
- **Resilient notifications** — office and agent emails are configured to continue the workflow even if the email step fails, so a notification hiccup never blocks the caller's booking or listing.

11. Setup Checklist

- Connect a Google Sheets credential and point every Sheets node at your own spreadsheet, with Listings, Viewings and Escalations tabs.
- Connect a Gmail credential and set the office/agent notification address on the three “Send ...” email nodes.
- Activate the workflow, then copy each webhook’s Production URL into the matching Retell custom tool: `check_property_availability`, `list_property`, `book_property_visit`, `get_property_details`, `request_human_callback`.
- Pre-populate the Listings sheet with a few sample properties (or approve some pending_review submissions) so availability search and property-detail lookups have real inventory to test against.
- Test all five tools end-to-end with a live or simulated call: search, list a property, book a viewing, ask for details, and request a callback.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating a voice-agent backend for real estate with inventory search, slot-conflict checking, approval-gated listings and human escalation.